

Activity 6: AI Project Discovery Challenge – SmartBank

Purpose

The objective of this activity is to apply the concepts learned throughout Day 1 to analyze a real-world software project using Generative AI. Learners will use prompt engineering techniques to generate software engineering artifacts, evaluate AI-generated outputs, and prepare a Project Discovery Document that can be referenced throughout the remaining training.

Pre-requisites

- Completion of Activities 1–5
 - ChatGPT or Oracle Chat AI
 - Internet connection
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Estimated Duration

45 Minutes

Scenario

Your Project Manager has scheduled a project kickoff meeting for the SmartBank Net Banking Application.

Before development begins, every development team is required to submit a **Project Discovery Document** that provides a high-level understanding of the project.

Rather than preparing the document manually, your team will use Generative AI to generate the initial draft. As software engineers, you are responsible for reviewing, refining, and validating the generated content before submission.

Tasks

Complete the following tasks in sequence.

Task 1: Create a Project Discovery Document

Use AI to generate a comprehensive overview of the SmartBank application.

Prompt

Act as a Senior Solution Architect.

Prepare a Project Discovery Document for the SmartBank Net Banking Application.

Include:

- Project Overview
- Business Objectives
- Scope
- Primary Users
- Key Features
- Major Functional Modules

Present the information using headings and bullet points.

Task 2: Generate a High-Level Architecture

Prompt

Based on the SmartBank application, describe the high-level system architecture.

Include:

- Frontend
- Backend
- Database
- External Systems
- APIs

Describe how these components interact.

Do not generate diagrams.

Task 3: Generate Risks and Assumptions

Prompt

Act as an Enterprise Solution Architect.

Identify possible assumptions and project risks for the SmartBank application.

Present the information in two sections:

- Assumptions
- Risks

Also suggest one mitigation strategy for each identified risk.

Task 4: Generate Questions for the Client

One of the most overlooked skills in software projects is asking the **right questions**.

Prompt

Act as a Business Analyst.

Based on the SmartBank project requirements, generate at least 15 clarification questions that should be asked to the client before development begins.

Group the questions into:

- Functional Requirements
 - Security
 - Performance
 - User Experience
 - Compliance
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Task 5: Validate the AI Response

Review everything generated in the previous tasks.

Identify:

- Information that is clearly supported by the project description.
- Information that appears to be an assumption.
- Areas requiring clarification from the client.
- Any hallucinations or unsupported statements.

Update the generated document accordingly.

Expected Deliverables

By the end of this activity, you should have:

- Project Discovery Document
 - High-Level Architecture Description
 - Assumptions and Risks Register
 - Client Clarification Questions
 - Reviewed and Validated AI Outputs
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Challenge Task

Your instructor will assign one module of the SmartBank application (e.g., Login, Fund Transfer, Dashboard, Bill Payments, Beneficiary Management).

Using everything learned during Day 1, create **one master prompt** that generates a complete project analysis for your assigned module.

Your prompt should instruct the AI to generate:

- Functional Requirements
- Non-Functional Requirements
- User Stories
- Business Rules
- Validation Rules
- Acceptance Criteria
- Suggested REST APIs
- Possible Database Tables
- Risks and Assumptions

Review the AI-generated response, refine it where necessary, and save it for use during the coding activities in subsequent training days.